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Health Indicators and Descriptive Epidemiology

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Objectives

- ▶ Describe health indicators and the characteristics of good health indicators
- ▶ Introduce the following descriptive epidemiology categories:
 - ▶ Person
 - ▶ Place
 - ▶ Time
- ▶ Evaluate age-adjusted liver cancer incidence in the United States by person, place, and time

Supporting Readings

- ▶ *Principles of Epidemiology in Public Health Practice*, Third Edition. An Introduction to Applied Epidemiology and Biostatistics. Section 6: Descriptive Epidemiology. Centers for Disease Control and Prevention, Atlanta, GA.
- ▶ Available at <https://www.cdc.gov/ophss/csels/dsepd/ss1978/lesson1/section6.html>



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Health Indicators

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What Is a Health Indicator?—1

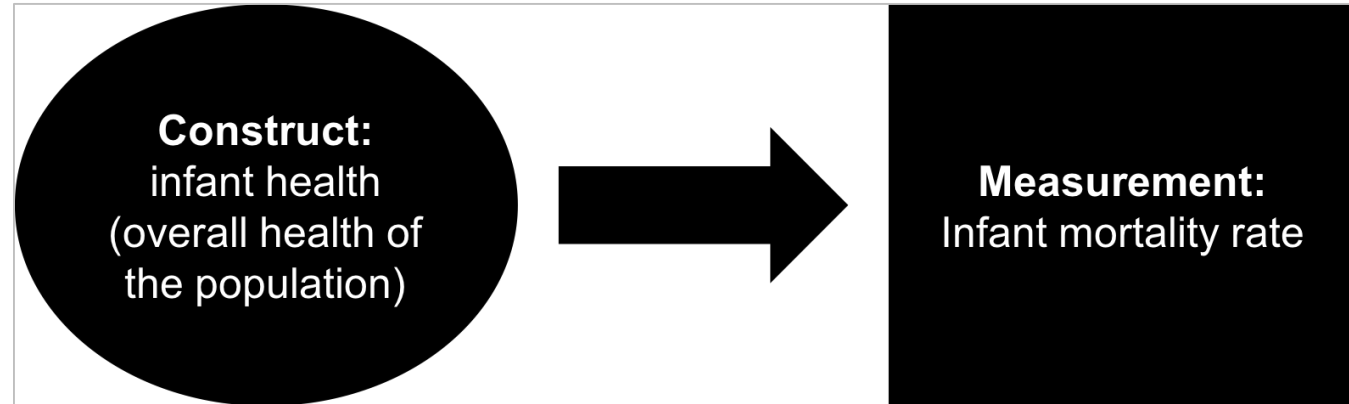
- ▶ Indicator = measure
- ▶ “A variable, susceptible to direct measurement, that reflects the state of health of persons in a community.”
 - ▶ Porta M. (2008), *A Dictionary of Epidemiology*, 5th edition, Oxford

What Is a Health Indicator?—2

- ▶ A health indicator “... defines a measure of health (i.e., the occurrence of a disease or other health-related event) ... among a specified population”
 - ▶ Lengerich EJ (ed.), 1999, *Indicators for Chronic Disease Surveillance: Consensus of CSTE, ASTCDPD, and CDC*. Atlanta, GA: Council of States and Territorial Epidemiologist
- ▶ “ ... these measures attempt to reflect and monitor the health status of a population”
 - ▶ PAHO, 2001, Health indicators: building blocks for health situation analysis. *Epi Bulletin PAHO*;22(4):1–5.

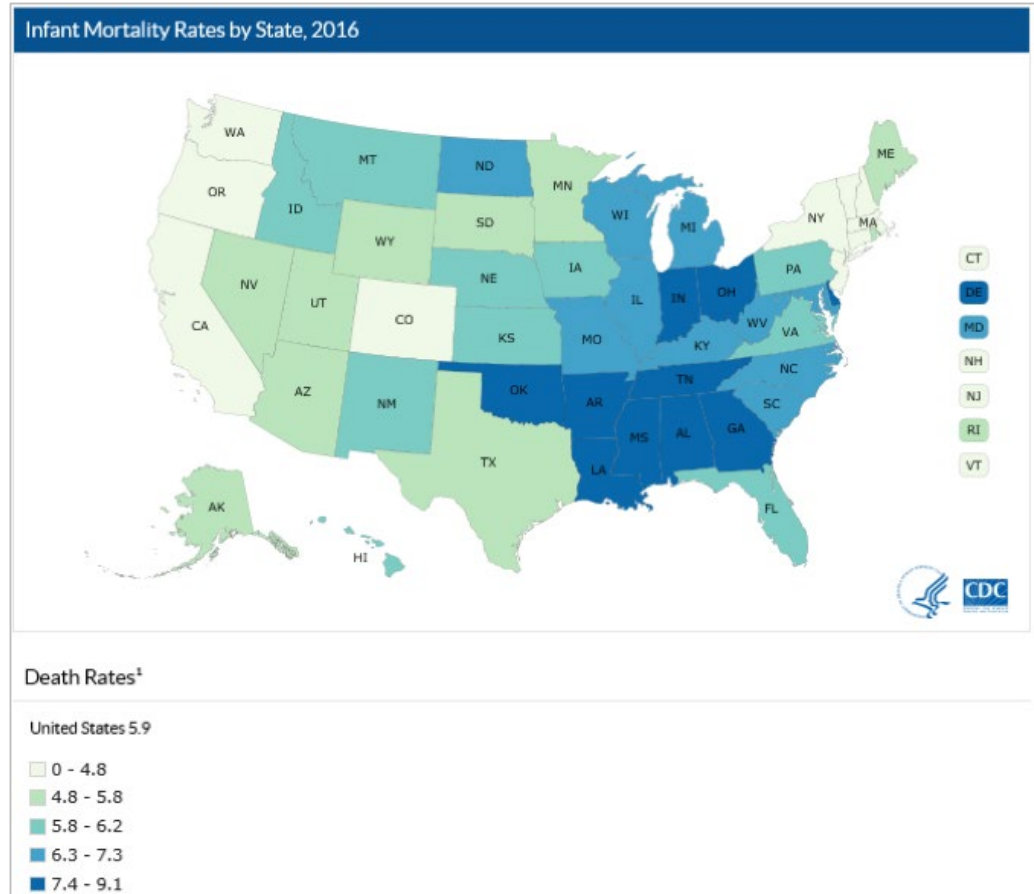
Health indicators: Measurement of a Construct

- ▶ Health indicators are not only a directly measured variable, but they represent a construct of health



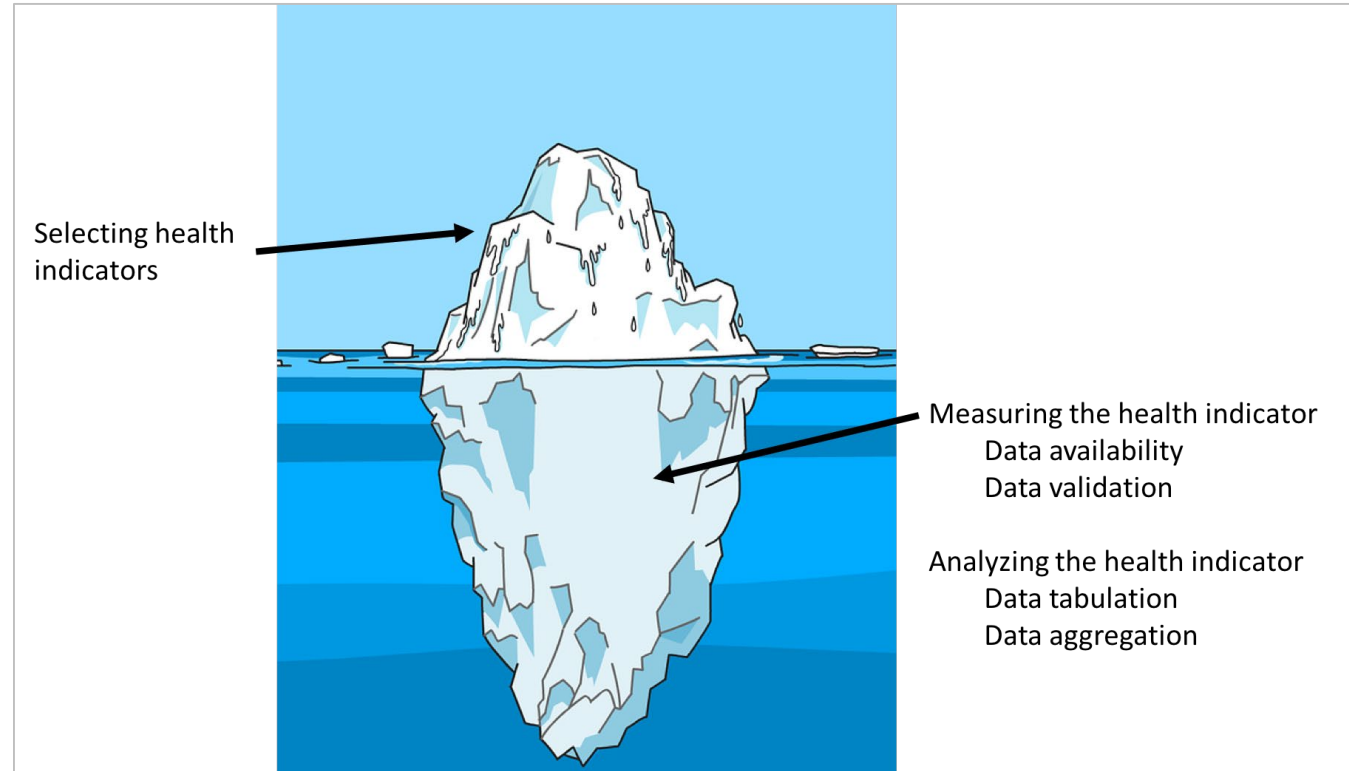
Health

Indicator: Infant Mortality Rate



Source: National Center for Health Statistics. Data source: CDC WONDER. Available at https://www.cdc.gov/nchs/pressroom/sosmap/infant_mortality_rates/infant_mortality.htm

Measurement of a Health Indicator



Health Indicator: Availability

- ▶ **Availability:** primary sources of data for health indicators

1. Public health information systems

- ▶ Surveillance systems
- ▶ Surveys of health outcomes and determinants
- ▶ Registries of health outcomes

2. Routine health information systems

- ▶ Vital registration systems
- ▶ Medical records
- ▶ Administrative databases

Health Indicator: Validation

- ▶ **Validation:** existing frameworks to determine data validity
- 1. Performance of Routine Information System Management (PRISM)
 - ▶ Tools to evaluate and strengthen routine health information systems
 - ▶ PRISM: <https://www.measureevaluation.org/resources/publications/ms-11-46-d>
- 2. Routine Health Information Network (RHINO)
 - ▶ Network of people working to improve routine health information systems
 - Data quality audit tool
 - Routine data quality assessment tool
 - ▶ RHINO: <http://rhinonet.org/index.php>

Health Indicator Measurement: Challenges to Availability and Validation

- ▶ Data availability and validation are challenges in many countries
 - ▶ Bureaucratic infrastructure
 - ▶ Political environment
 - ▶ Economic situation
 - ▶ Sufficient skilled work force

Health Indicator Analyses: Data Tabulations

- ▶ Standard definitions for indicators are important for the data to tell the same truthful story in the population(s)
- ▶ Indicators must be measured, or tabulated, consistently
 - ▶ Across time (comparing a population to itself)
 - Time trend analysis in a single population
 - ▶ Across populations (comparing separate populations)
 - Comparing the health indicator in ≥ 2 populations

Health Indicator Analyses: Data Aggregation

- ▶ Indicators with numerators and denominators must apply groupings to both the numerator and denominator
 - ▶ Aggregating groups by age, gender, race/ethnicity
- ▶ Individual-level data is not often available due to privacy concerns
 - ▶ If you are combining numerator data from source A with denominator data from source B to estimate the indicator, the data must be available in the same groups from each source
- ▶ Disclosure limitation standards
 - ▶ Data will not be disseminated in a specified grouping when there are $< x$ people in that group



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“I’ve written about how mortality is
a wonderful indicator of societal
progress.”

—Sir Angus Deaton,
Economist and author

Selecting Health Indicators

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Who Uses Health Indicators?

- ▶ Health indicators are incredibly powerful because they are:
 - ▶ Politically charged
 - Maternal and child health indicators are frequently interpreted as measures of how well a government is caring for the people
 - ▶ Scrutinized for funding needs
 - Non-governmental organizations (NGOs)
 - Private donors
 - Private sector
 - ▶ Interpreted for security purposes
 - US Central Intelligence Agency (CIA)

Important to Select a Strong Indicator

- ▶ Health indicators are the building blocks of epidemiology in public health practice
 - ▶ “Garbage in, garbage out” principle

How Do You Select a Health Indicator?

- ▶ Characteristics of strong health indicators
 - ▶ Availability
 - ▶ Quality
 - Validity
 - Tabulation
 - Aggregation
 - ▶ Utility
 - Easy to use and interpret
 - Understandable by information users (for example, managers, decision makers)
 - Policy relevance

Sources of Health Indicators

- ▶ World Health Organization (WHO) keeps a compendium and registry of key health indicators
- ▶ Pan American Health Organization (PAHO) as a set of Basic Health Indicators
- ▶ Institute of Medicine (IOM) has a committee that monitors the state of health indicators in the United States
- ▶ Specific health topic areas have health indicators
 - ▶ Nutrition
 - ▶ Aging



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“If you don’t know where you are
going any road can take you there.”

—Lewis Carroll,
Alice in Wonderland

Descriptive Epidemiology: Person, Place, and Time Trends

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What Is Descriptive Epidemiology?

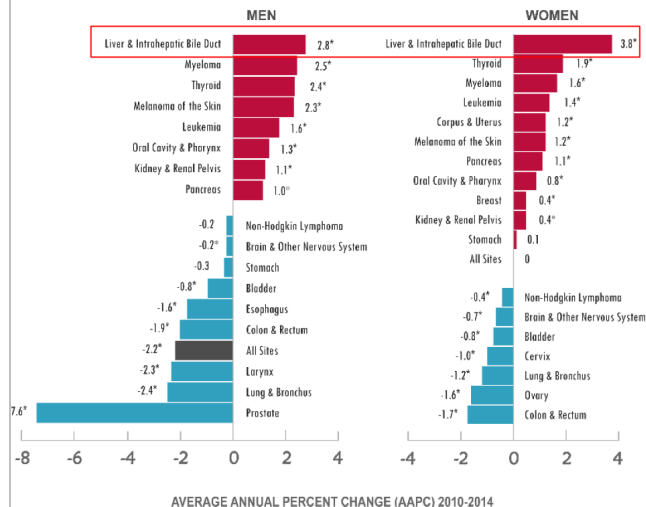
- ▶ Porta M., *A Dictionary of Epidemiology*, sixth edition, Oxford 2014:
 - ▶ “General descriptions concerning the relationship of disease to basic characteristics such as age, gender, ethnicity, occupation, social class, and geographic location; even such general descriptions may have analytic dimensions.”
 - ▶ “Major categories in descriptive epidemiology are
 - Persons
 - Place, and
 - Time” (trends)

Exercise: Cancer Incidence Trends in the United States

► Cancer incidence rate =

$$\frac{\text{Number of new cancers}}{\text{Number in the population}} \times 100,000$$

NATIONAL TRENDS IN RATES OF NEW CANCER CASES



*AAPC is significantly different from zero ($p < .05$).

Rates were adjusted for reporting delay in the registry.

seer.cancer.gov

Source: Annual Report to the Nation, Part 1: National Cancer Statistics

Source: Cancer Incidence Rates. National Cancer Institute Surveillance, Epidemiology and End Results Program. Available at: <https://seer.cancer.gov/statistics/types/incidence.html>



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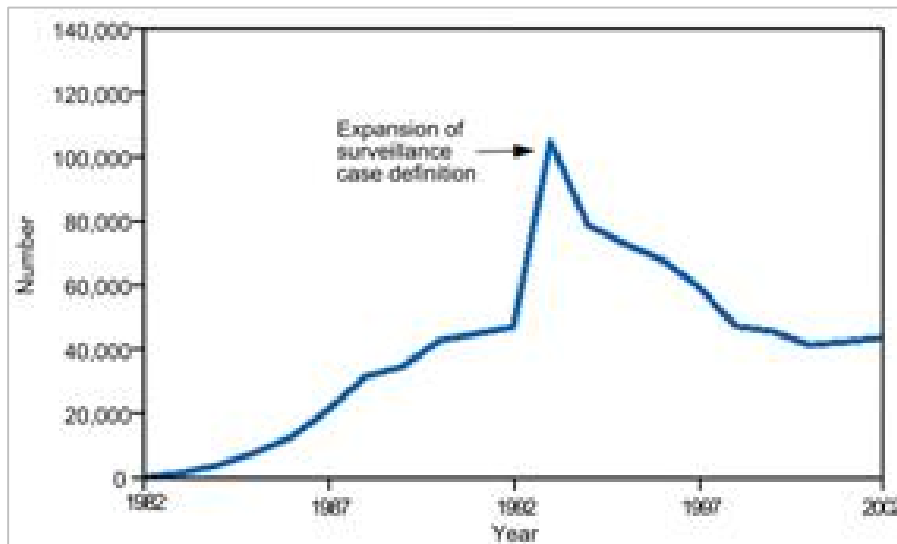
Descriptive Epidemiology: Time

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What Is a Time Trend?

- ▶ Describing the change in a health indicator over time
- ▶ How do you analyze a trend?
 - ▶ Plot (x axis is usually “time”)
- ▶ Challenges of a trend
 - ▶ Data hasn't been collected over time
 - ▶ Change in definition of the health indicator over time
 - ▶ Outbreaks

Reported cases of AIDS by year, United States and US territories, 1982–2002



Exercise: Age-Adjusted Liver Cancer Incidence in the United States

- ▶ CDC WONDER is an excellent source of aggregated health indicators
 - ▶ Age-adjusted liver cancer incidence rates and 95% confidence intervals
 - ▶ <https://wonder.cdc.gov/cancer-v2014.HTML>
- ▶ Liver cancer incidence rate
 - ▶ Cancer incidence rate =

$$\frac{\text{Number of new cancers}}{\text{Number in the population}} \times 100,000$$

- ▶ Direct age standardization using the 2000 US standard population

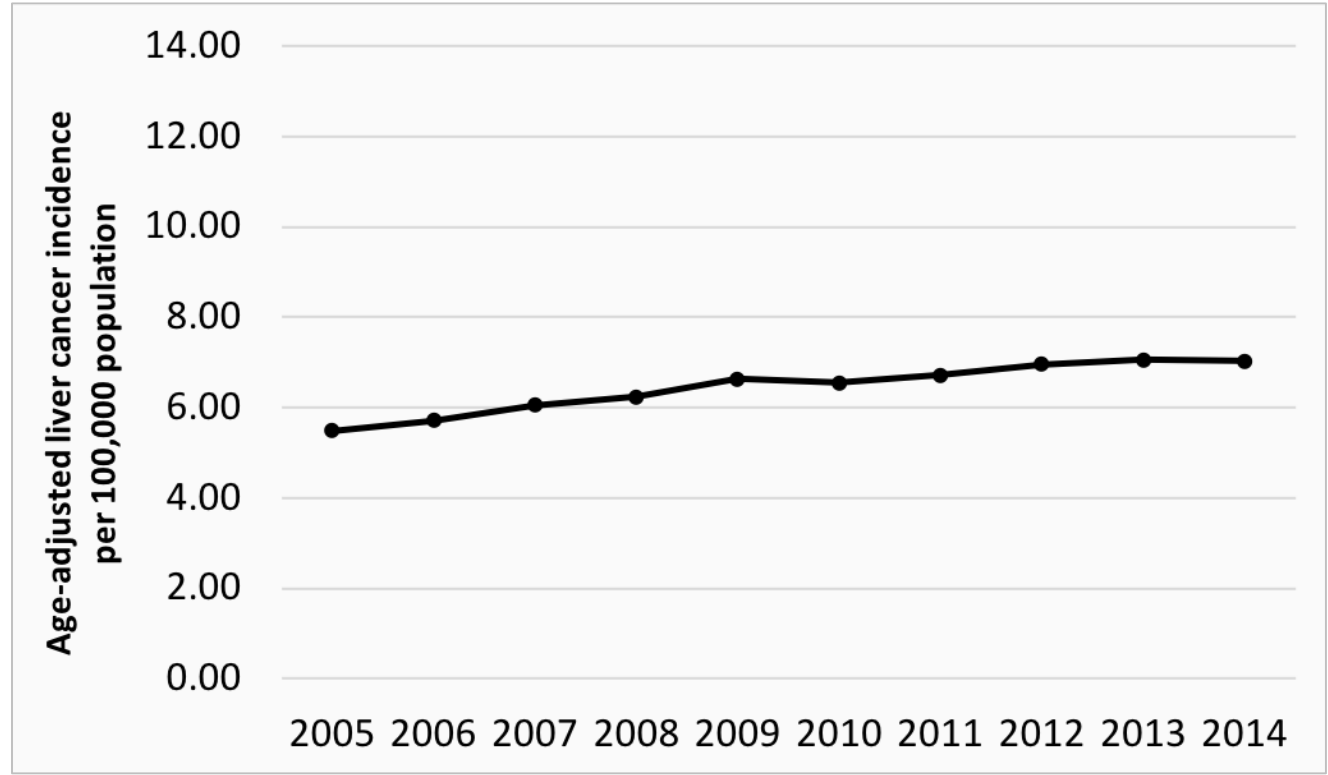
Work it Out: Problem

- ▶ **What is the trend in age-adjusted liver cancer incidence in the united states?**
- ▶ In each year from 2005–2014, graph the age-adjusted cancer incidence rate
- ▶ A marker for each year's estimate, connected by a line

Year	Age adjusted rate
2005	5.48
2006	5.71
2007	6.05
2008	6.24
2009	6.63
2010	6.54
2011	6.71
2012	6.95
2013	7.05
2014	7.03

Work it Out: Solution

Age-adjusted liver cancer incidence rates in the US, 2005–2014





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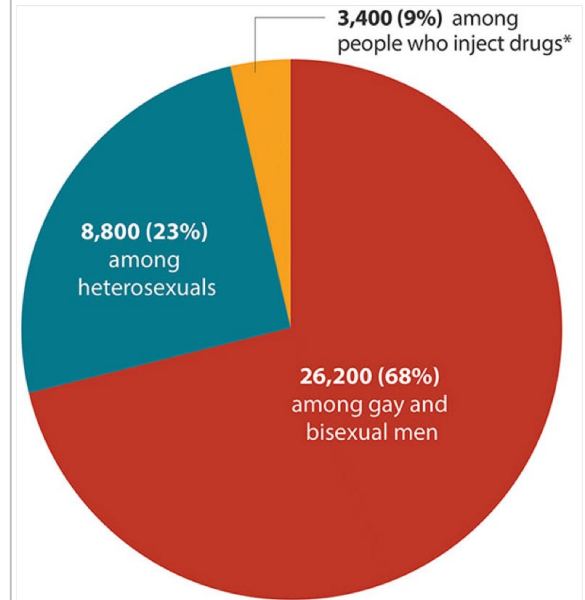
Descriptive Epidemiology: Person

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Person—1

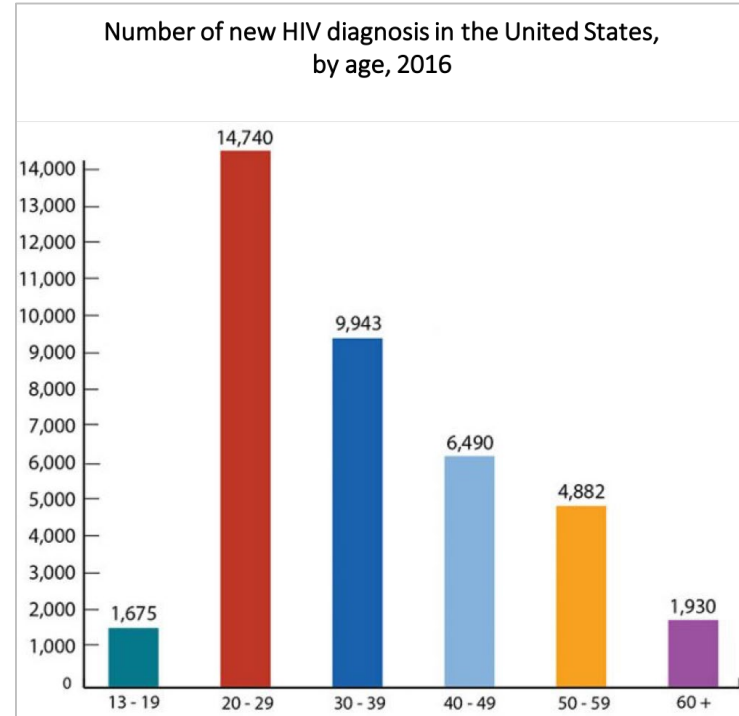
- ▶ Stratifying a health indicator by *person characteristics* can help identify differences in the health indicator among sub-populations
 - ▶ Basic demographics:
 - Age
 - Race/ethnicity
 - Sex
 - Socioeconomic status
 - ▶ Health indicator-specific person characteristics
 - Sexual orientation
 - Occupation

Estimated new HIV infections in the United States, by transmission category, 2015



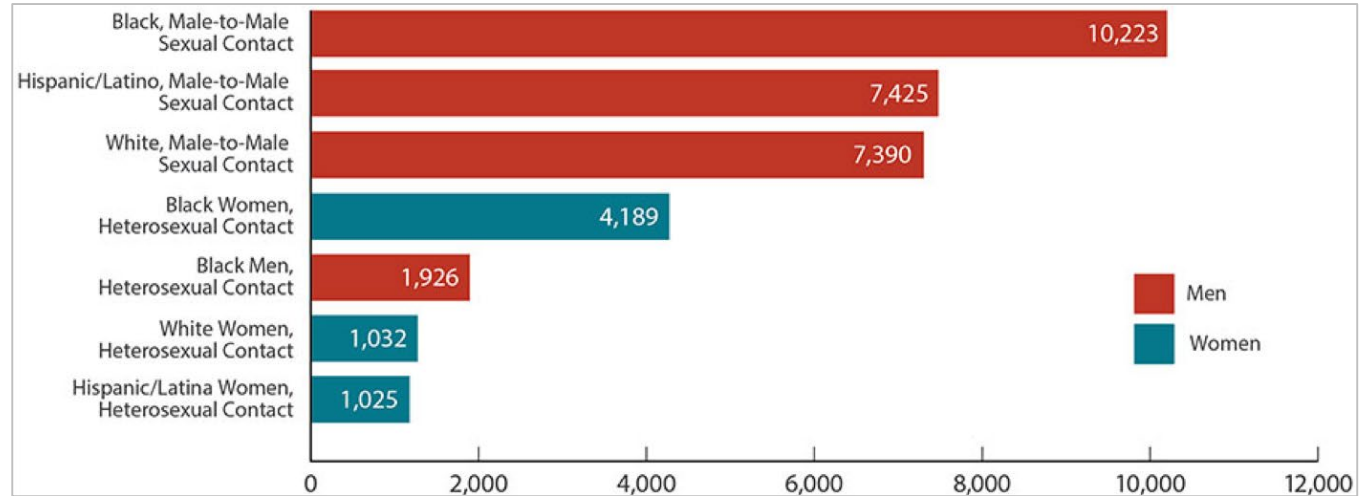
Person—2

- ▶ Stratifying a health indicator by *person* characteristic: age



Person—3

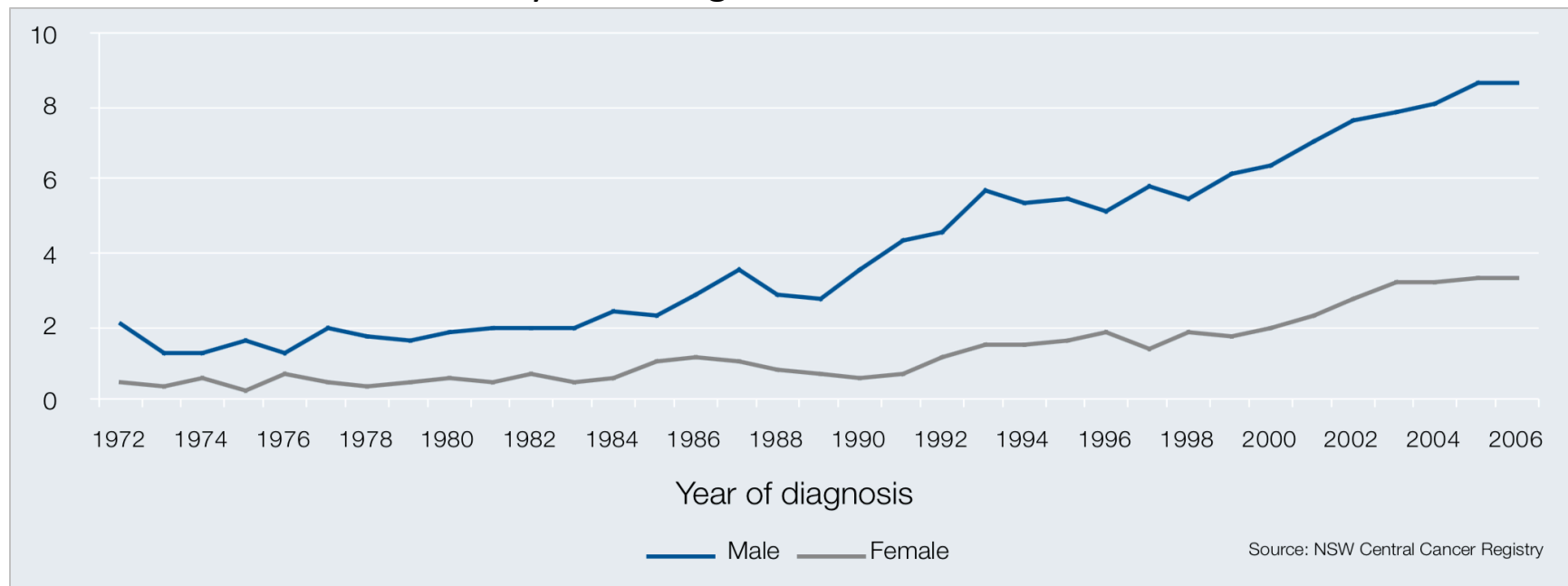
- ▶ Stratifying a health indicator by more than one person characteristic:



Source: CDC. Diagnoses of HIV infection in the United States and dependent areas, 2016. *HIV Surveillance Report* 2017;28.

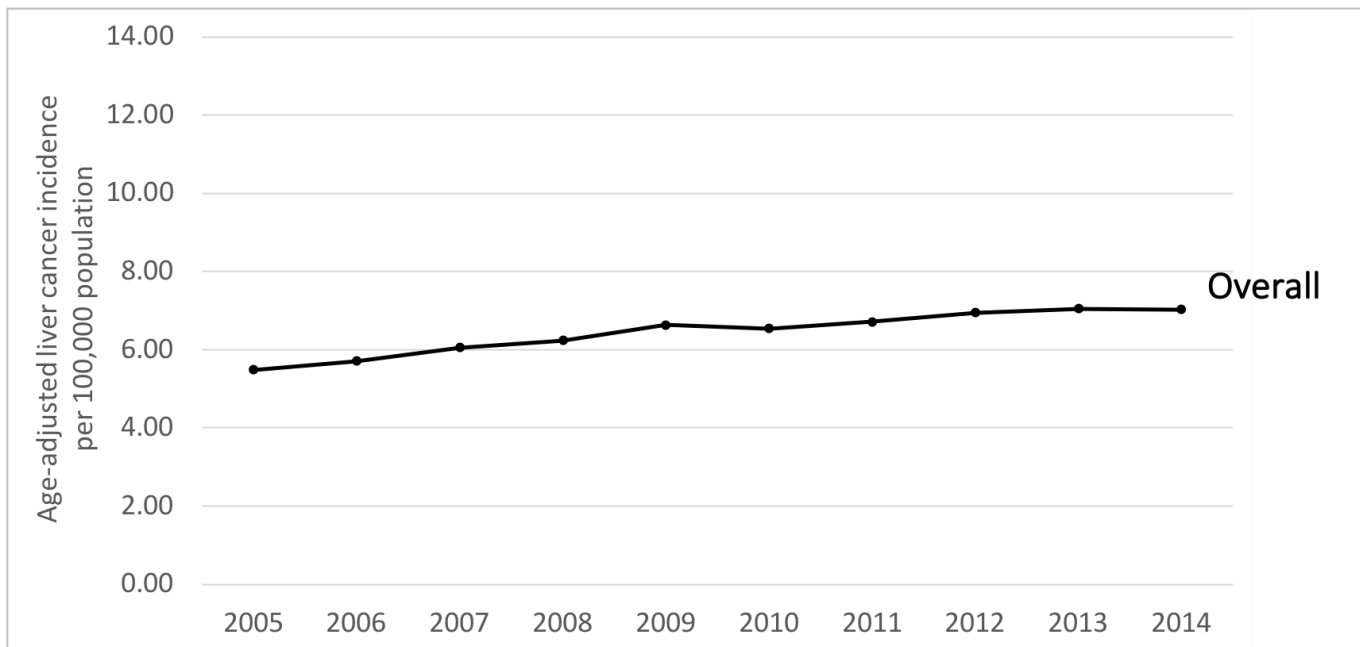
Trends in Liver Cancer in New South Wales (NSW), Australia

Trends in liver cancer: age-standardized incidence rate per 100,000 population by sex, all ages, NSW 1972–2006



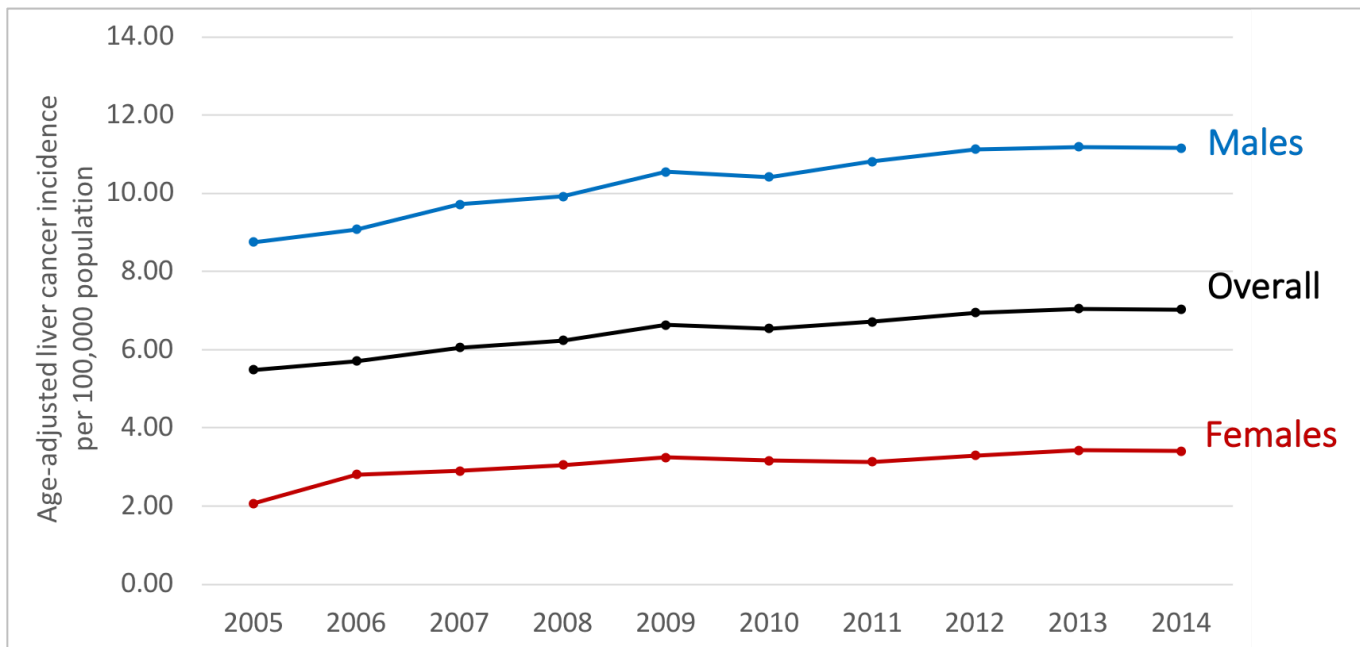
Work it Out: Male and Female?

Age-adjusted liver cancer incidence rates in the US, 2005–2014



Work it Out: Solution

Age-adjusted liver cancer incidence rates in the US, by sex, 2005–2014





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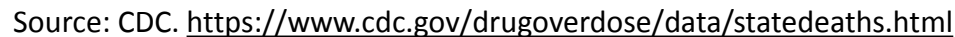
Descriptive Epidemiology: Place

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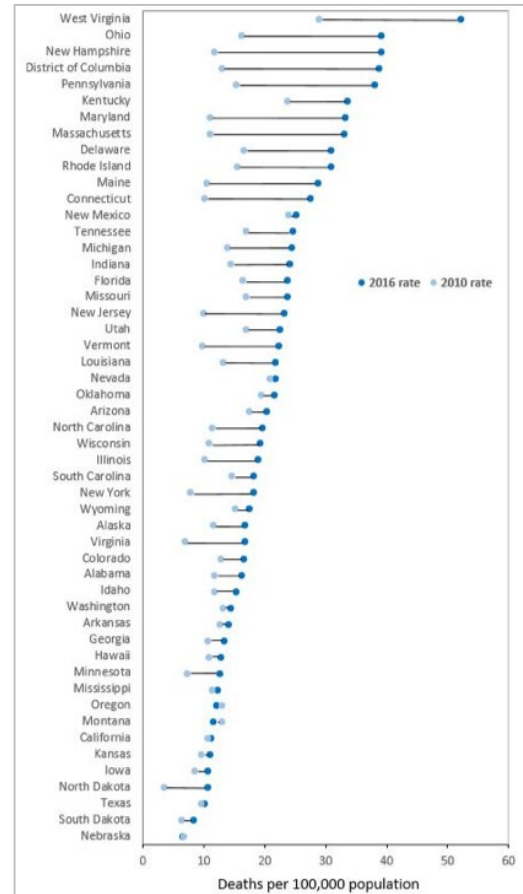
What Is *Place*?

- ▶ Stratifying a health indicator *by place* can help identify differences by geography or other characteristics of place
 - ▶ Basic place characteristics
 - Geographical
 - ▶ Other place characteristics
 - Cultural
 - Social
 - Political
 - Economic
 - Historical
 - Ethnic

Number and age-adjusted rates of drug overdose deaths by state,
US 2016

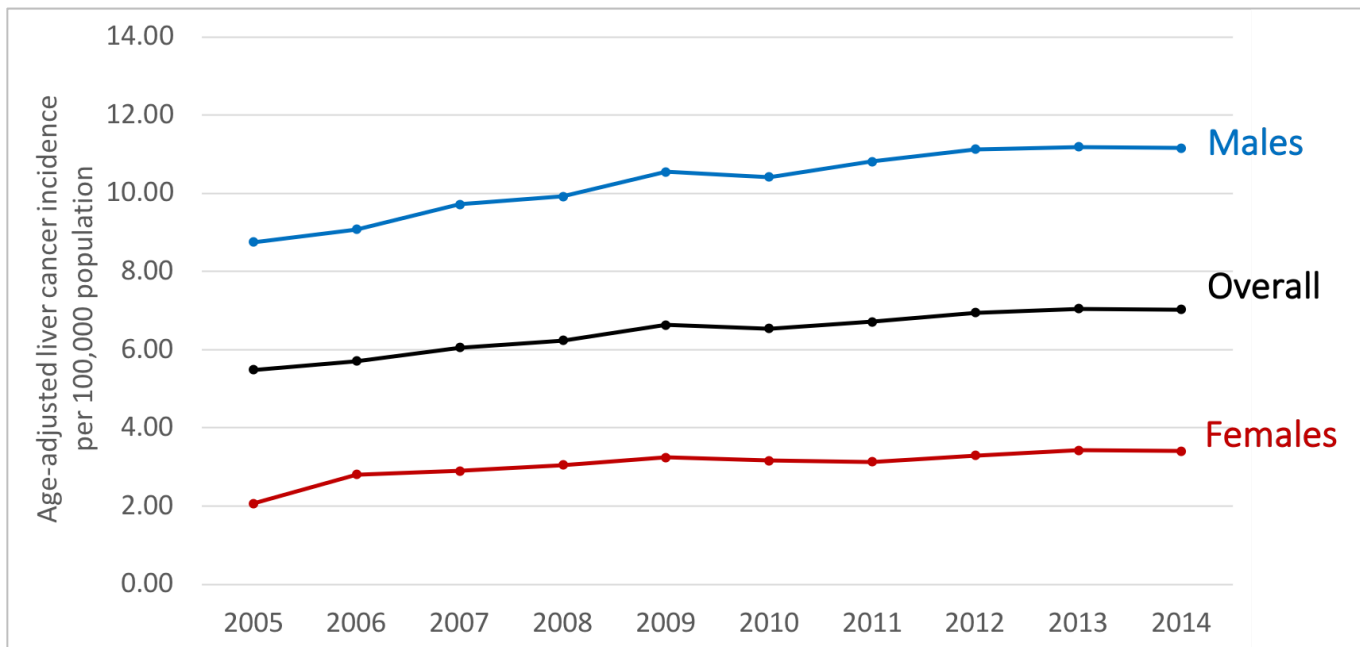


Age-adjusted rate of drug overdose deaths, by state, 2010, 2016



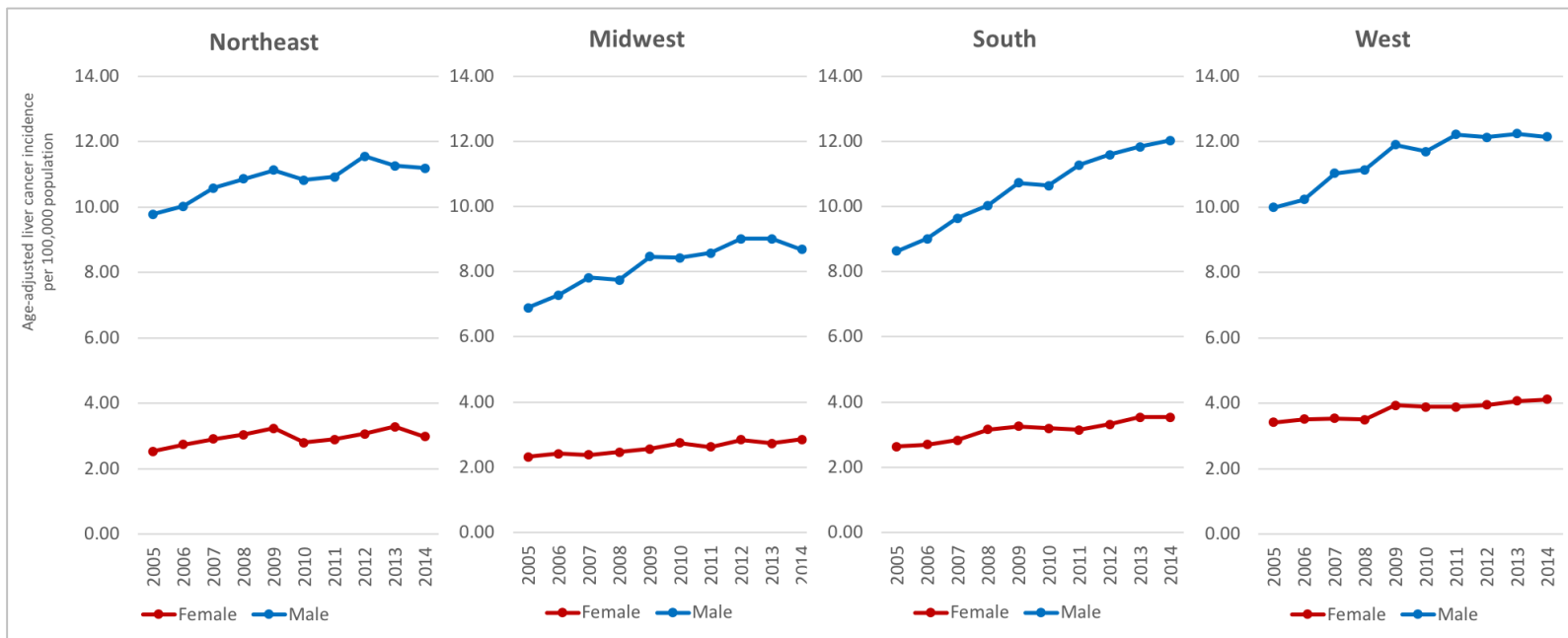
Work it Out: Plot for each Region?

Age-adjusted liver cancer incidence rates in the US, by sex, 2005–2014



Work it Out: Solution

Age-adjusted liver cancer incidence rates in the US, by region and sex, 2005–2014





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Summary

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Lecture Summary

- ▶ Health indicators are the foundation of epidemiology in public health practice
 - ▶ Construct and measurement
 - ▶ Availability, validity, tabulation, aggregation
 - ▶ Many validate indicators exist
- ▶ Descriptive epidemiologic analyses use health indicators to better understand the problem
 - ▶ Person, place, time
 - ▶ Combinations